

CERT/CC Vulnerability Note VU#706695

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Content

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VU#706695 - Checkbox Survey insecurely deserializes ASP.NET View State data

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Software Engineering Institute

CERT Coordination Center

Checkbox Survey insecurely deserializes ASP.NET View State data

Vulnerability Note VU#706695

Original Release Date: 2021-05-25 | Last Revised: 2021-05-25

Overview

Checkbox Survey prior to version 7.0 insecurely deserializes ASP.NET [View State](#) data, which can allow a remote, unauthenticated attacker to execute arbitrary code on a vulnerable server.

Description

CVE-2021-27852 Checkbox Survey insecurely deserializes ASP.NET View State data.

[Checkbox Survey](#) is an ASP.NET application that can add survey functionality to a website. Prior to version 7.0, Checkbox Survey implements its own View State functionality by

accepting a `_VSTATE` argument, which it then deserializes using [LosFormatter](#). Because this data is manually handled by the Checkbox Survey code, the ASP.NET ViewState Message Authentication Code (MAC) setting on the server is ignored. Without MAC, an attacker can create arbitrary data that will be deserialized, resulting in arbitrary code execution.

This vulnerability is reportedly being exploited in the wild.

Impact

By making a specially-crafted request to a server that uses Checkbox Survey 6.x or earlier, a remote, unauthenticated attacker may be able to execute arbitrary code with the privileges of the web server.

Solution

Apply an update

Starting with Checkbox Survey 7.0, View State data is not used. Therefore, Checkbox Survey versions 7.0 and later do not contain this vulnerability.

Remove Checkbox Survey versions older than 7

Checkbox is no longer developing Checkbox Survey version 6, so this version is no longer safe to use. If you are unable to update to an unaffected version of Checkbox Survey, this software should be removed from any systems that have it installed.

Acknowledgements

Thanks to the reporter who wishes to remain anonymous.

This document was written by Will Dormann.

Vendor Information

706695

Filter by status: All Affected Not Affected Unknown

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References

- <https://www.checkbox.com/>
- [https://docs.microsoft.com/en-us/previous-versions/aspnet/bb386448\(v=vs.100\)](https://docs.microsoft.com/en-us/previous-versions/aspnet/bb386448(v=vs.100))
- <https://docs.microsoft.com/en-us/dotnet/api/system.web.ui.losformatter?view=netframework-4.8>
- <https://swapneildash.medium.com/deep-dive-into-net-viewstate-deserialization-and-its-exploitation-54bf5b788817>

Other Information

| **CVE IDs:** | [CVE-2021-27852](#) | | | **Date Public:** | 2021-05-25 | | | **Date First Published:** | 2021-05-25 | | | **Date Last Updated: ** | 2021-05-25 20:33 UTC | | | **Document Revision: ** | 1 | |

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